

Contionuous VS Discrete PID Controller

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Abstract—As PID control algorithm is by far the most important and commonly applied control algorithm. As a consequence, it is very important for all the engineering students to have a good understanding of it. During the study of Control theory, the students usually get well acquainted with continuous systems, because they know the underlying mathematics quite well. In general, the courses dealing with discrete mathematics are not so common. So many students have never heard of discrete equations. In this paper a possible approach for addressing this problem is presented. A PID controller is realized both as continuous and discrete system. Their performance is compared on a real example.

Keywords—PID control algorithm; continuous system; discrete system; DC motor

I. INTRODUCTION

A proportional–integral–derivative (PID) is by far the most important control algorithm. Its beginnings can be traced back to the first half of the twentieth century, especially in the field of process industry characterized by processes which are not only highly complex, but also nonlinear and subject to long time delays [1]. In process control, more than 95 percent of the controllers are of PID type, where the majority of them are actually of PI type [2]. In the whole market of controllers, the PID controller now represents more than ninety percent [3]. This widespread application of PID controllers can be attributed to their simplicity in addition to good performance which they are able to provide for a wide range of processes and excellent robustness to uncertainties [4]. Consequently, they have become the standard controller in industrial settings. It can be found in various kinds of controller implementations, either as a stand-alone (single-station) controllers or as a functional block in Programmable Logic Controllers (PLCs) and Distributed Control Systems (DCSs) [5].

Despite of its relative maturity the field of PID control is still an area of vivid research. As the structure of the PID controllers is given, the main emphasis is given to the correct selection of the PID control algorithm's parameters (controller tuning). Despite the fact that there are 408 various sources of PID controller tuning methods [6] only one fifth of controllers in production environment are properly tuned [7]. Actually, 30% of controllers permanently operate in manual mode and 25% use factory tuning without any update with respect to the given plant [8]. PID controllers are still

very often applied in research papers in various scientific areas and might even be the main topic of research [9]–[18]. Therefore, it is clear that a good background in the PID control theory and application is needed for students in order that they are able to transfer that knowledge to the production environment.

In this paper an approach dealing with this problem is presented. Both continuous as well as discrete approach are dealt with. The controllers are namely first met in Feedback Control Theory courses which are based on continuous signals. In practice, the PID control algorithms are commonly implemented in microcontroller which of course operate with digital signals.

II. PID CONTROL ALGORITHM

The proportional–integral–derivative (PID) control algorithm is defined by the following transfer function:

$$P(s) = K_p + \frac{K_I}{s} + K_D s \quad (1)$$

It can be seen that it is made of three parallel components (proportional, integral, and derivative). That's where its name comes from. Its block diagram is shown in Fig. 1.

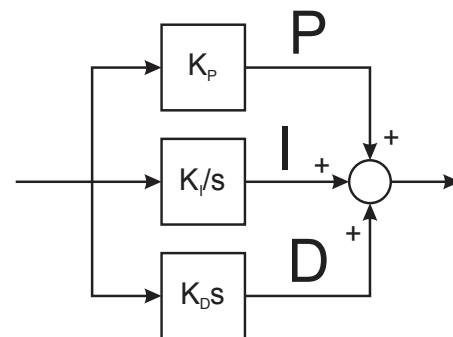


Figure 1. Block diagram of a PID controller.

The PID control algorithm can for example be changed to PD if only the first and the third component are present. The transfer function in (1) is often rewritten in the following equivalent form:

$$P(s) = K_p \left(1 + \frac{1}{T_I s} + T_D s \right) \quad (2)$$

III. DESIGN OF PID CONTROLLER

A. Continuous Type

The main building block for the design of the (continuous) PID controller is the operational amplifier shown in Fig. 2.

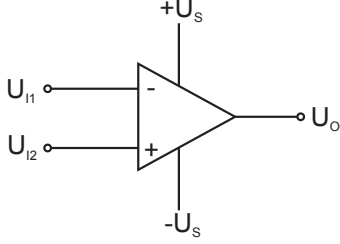


Figure 2. Operational amplifier.

The inputs $+U_S$ and $-U_S$ are attributed to the supply. The output U_O is related to the input voltages by the following equation:

$$U_O = A(U_{I2} - U_{I1}) \quad (3)$$

The parameter A is very large, so when doing modeling it is usually supposed to be infinite. The first element needed to construct PID controller is the inverting amplifier shown in Fig. 3.

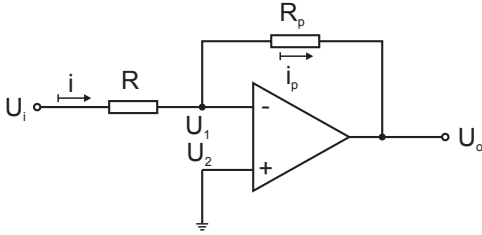


Figure 3. Inverting amplifier.

Its transfer function can be obtained by assumptions that $U_1=U_2=0$ and that the current through the amplifier is negligible ($i=i_p$). This gives us:

$$\frac{U_i - 0}{R} = \frac{0 - U_O}{R_p} \Rightarrow U_O = -\frac{R_p}{R} U_i \quad (4)$$

So, at least in theory the any amplification can be obtained by a proper combination of R_p and R . If we neglect the minus sign, we can realize proportional component by inverting amplifier. A slight modified amplifier can be used to obtain integral component (see Fig. 4).

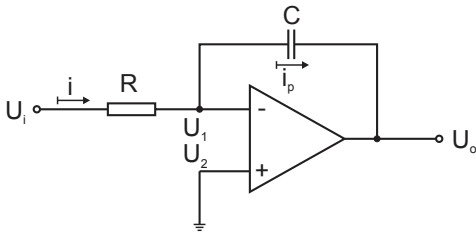


Figure 4. Integrating amplifier.

Its transfer function is obtained by:

$$\frac{U_i - 0}{R} = \frac{0 - U_O}{\frac{1}{Cs}} \Rightarrow U_O = -\frac{1}{RCs} U_i \quad (5)$$

So, in this way the integral component can be obtained. If we exchange the roles of resistor and capacitor we get the derivative component. In order to sum them, we need the summing amplifier shown in Fig. 5.

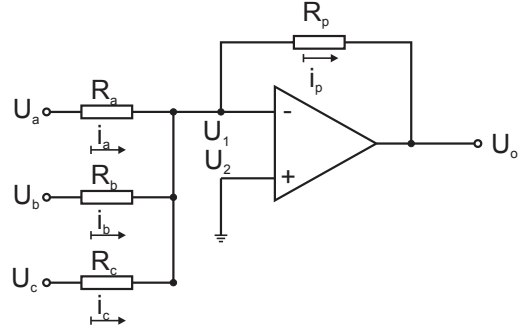


Figure 5. Summing amplifier.

Its transfer can be obtained, if we assume $i_a + i_b + i_c = i_p$. This gives us:

$$\frac{U_a - 0}{R_a} + \frac{U_b - 0}{R_b} + \frac{U_c - 0}{R_c} = \frac{0 - U_O}{R_p} \Rightarrow U_O = -\left(\frac{R_p}{R_a} U_a + \frac{R_p}{R_b} U_b + \frac{R_p}{R_c} U_c\right) \quad (6)$$

So, if we combine inverting amplifier, integrating amplifier, differentiating amplifier together with the summing amplifier, we get the structure, shown in Fig. 6.

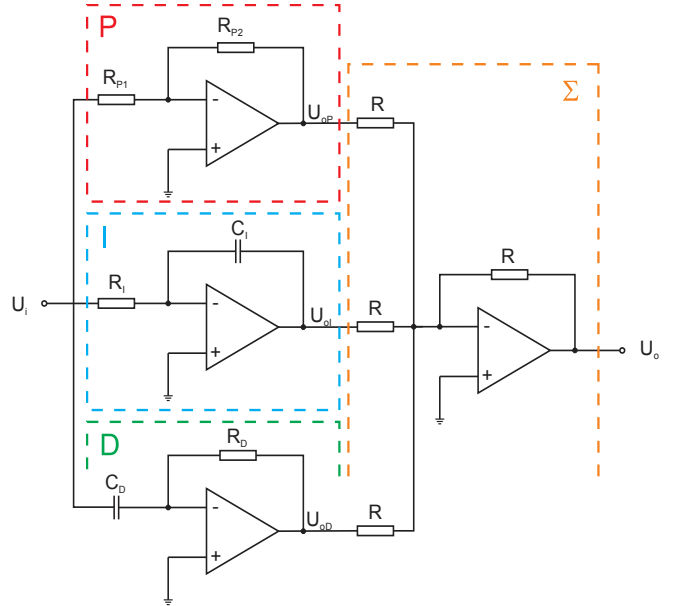


Figure 6. (Continuous) PID controller.

In this way, we can get a fully functional PID controller. We should however be aware of the fact, that its output voltage is limited with the supply voltage (approximately)

which is around $\pm 15V$. The realized version of the PID controller is shown in Fig. 7.

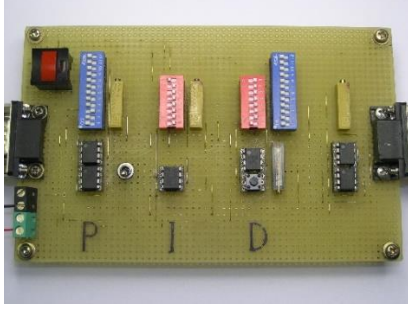


Figure 7. Realized (continuous) PID controller.

B. Discrete Type

Discrete type PID controller can easily be implemented using any type of microcontroller. We have used PIC microcontroller (an example is shown in Fig. 8).

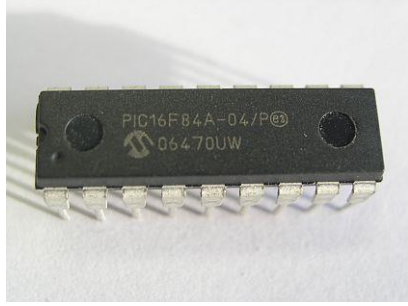


Figure 8. Microcontroller PIC16F84A.

The most comfortable way of using microcontrollers, at least at the beginner's level, is via a development board (an example is shown in Fig. 9).



Figure 9. EasyPIC4 development board next to a notebook.

In this way, it is easy to program the microcontroller and then test its performance prior to actual application in the production environment.

In general, students have no problem with the concept of discrete signal $x[kT_s]$. It is a signal, the value of which is known only for a sequence $T_s, 2T_s, 3T_s, \dots$, where T_s is the sampling time. If $e(t)$ is the error signal and $o(t)$ the output of a PID controller, we can write:

$$o(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt} = o_P(t) + o_I(t) + o_D(t) \quad (7)$$

So, for the discrete proportional component we can easily get:

$$o_P[kT_s] = K_P e[kT_s] \quad (8)$$

The integral component can be evaluated based on Fig. 10.

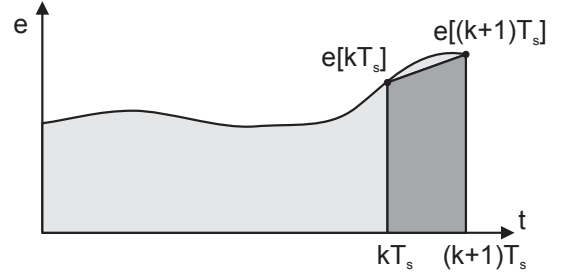


Figure 10. Discrete integration.

We can write:

$$o_I[(k+1)T_s] \cong o_I[kT_s] + K_I \frac{T_s}{2} (e[(k+1)T_s] + e[kT_s]) \quad (9)$$

The derivative component can be obtained using the fact that integration and differentiation are the inverse operations. So, for the differentiation (9) can be rewritten into:

$$e[(k+1)T_s] \cong e[kT_s] + K \frac{T_s}{2} (o_D[(k+1)T_s] + o_D[kT_s]) \quad (10)$$

This can be rewritten to:

$$o_D[(k+1)T_s] + o_D[kT_s] \cong K_D \frac{2}{T_s} (e[(k+1)T_s] - e[kT_s]) \quad (11)$$

In order to simplify the notation, the so-called z-operator is employed. It is defined by:

$$O(z) = o[kT_s] \Rightarrow zO(z) = o[(k+1)T_s] \quad (12)$$

Eq. 9 can now be modified to the following form:

$$zO_I(z) \cong O_I(z) + K_I \frac{T_s}{2} (zE(z) + E(z)) \quad (13)$$

This finally gives us:

$$O_I(z) \cong K_I \frac{T_s}{2} \frac{z+1}{z-1} E(z) \quad (13)$$

In a similar way, we can analyze the derivative component. In this case, we get the following result:

$$O_D(z) \cong K_D \frac{2}{T_s} \frac{z-1}{z+1} E(z) \quad (14)$$

So, in the end, we get the discrete transfer function for a PID controller can be expressed as:

$$P(z) \cong \left(K_P + K_I \frac{T_s z + 1}{2 z - 1} + K_D \frac{2 z - 1}{T_s z + 1} \right) E(z) \quad (15)$$

This is still not the form that can be used in a microcontroller. It can however be used to compare the performance of a continuous and discrete PID controller. This can easily be done in Matlab. If we for example take the transfer function $P(s)=2(1+1/100s+3s)$ we can get to $P(z)$ using the following analogy:

$$\frac{2 z - 1}{T_s z + 1} \sim s \quad (16)$$

It can be obtained by comparing (15) and (1). For the above mentioned transfer function, and the assumption that $T_s=0.01s$, we get:

$$P(s) = 2 \left(1 + \frac{1}{100s} + 3s \right) = \frac{600s^2 + 200s + 2}{100s} \quad (17)$$

Using the analogy (16) we get the discrete transfer function:

$$P(z) = \frac{600 \left(\frac{2 z - 1}{0.01 z + 1} \right)^2 + 200 \left(\frac{2 z - 1}{0.01 z + 1} \right) + 2}{100 \left(\frac{2 z - 1}{0.01 z + 1} \right)} \approx \frac{1202 z^2 - 2400 z + 1198}{z^2 - 1} \quad (18)$$

It can also be easily obtained using Matlab. We should type the following code:

```
num=[600 200 2];
denum=[100 0];
TF-cont=tf(num,denum);
TF-disc=c2d(TF-cont,0.01,'t')
```

From (18) we get the following discrete equation:

$$o[(k+2)T_s] = o[kT_s] + 1202e[(k+2)T_s] - 2400e[(k+1)T_s] + 1198e[kT_s] \quad (19)$$

It can easily be programmed onto a microcontroller using a development board (see Fig. 9). After the performance is verified, the microcontroller is put onto another board (see Fig. 11).

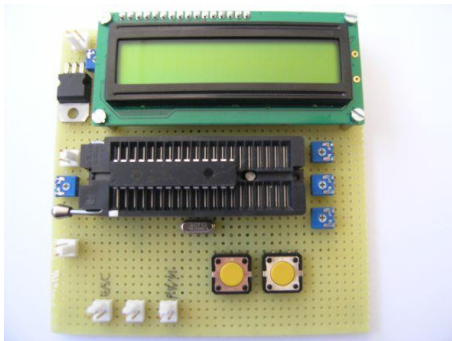


Figure 11. Discrete PID controller.

This is the one used to control the object.

IV. EXPERIMENTAL VERIFICATION

In order to verify the performance of both continuous and discrete PID controllers, they are used in an actual closed loop control system. In our case, it is a DC motor shown in Fig. 12.

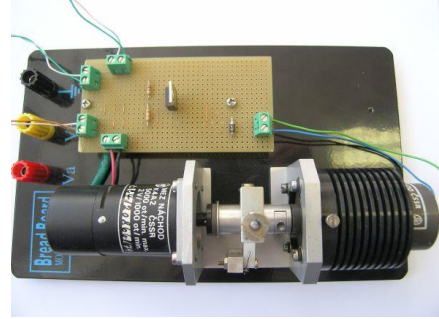


Figure 12. EasyPIC4 development board next to a notebook.

The block diagram of the whole close-loop control system is shown in Fig. 13.

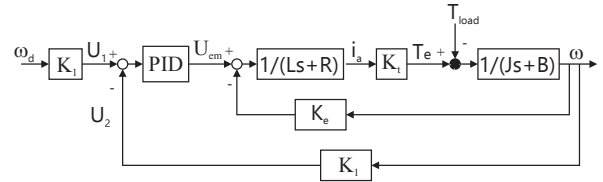


Figure 13. EasyPIC4 development board next to a notebook.

The PID controller shown in the block diagram can either be continuous or discrete.

V. CONCLUSION

The experimental setup mentioned above is used in lab experimental work at the University of Ljubljana. As the students lack sufficient knowledge in discrete systems, the presented experiment was very well received among them.

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